Francesco Zoffoli

Software engineer, public speaker and book author with experience in leading teams, improving their engineering culture, and delivering high quality products. He is an expert in C++, Python and the technologies used in the modern back-end stack, has in-depth knowledge in designing, implementing and growing distributed systems, and is passionate about efficient, reliable and simple software.

Contacts

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Work Experience

Present – Apr 2020 — Meta Inc., London & Remote (Italy)
Staff Software Engineer

I worked as staff engineer on the detection systems used to identify production issues happening in Facebook, Whatsapp, Instagram and all the Meta's systems. I operated across 4 teams (30 people), leading the technical strategy and following in the implementation of complex systems enabling billions in prevented revenue loss. I contributed to set the vision of the detection products of the group, I coordinated external relationships with partner teams and grew the teams through mentoring and hiring. I also helped drive department wide initiatives, grew the scope of the teams, and brought a change in the culture of the organization around reliability excellence and software quality.
My contributions included

- redefining the approach to monitoring by leveraging SLOs inside Meta, including defining the product functionality, the design of the system, and driving the adoption across different organizations, resulting in 10x more effective monitoring.
- driving the re-architecture of the existing alerting system into a highly available and reliable design, enabling the system to scale horizontally, growing 100x with the same hardware utilization and reducing outages by 50x.
- setting up a continuous program leveraging AI to identify and address gaps in the monitoring strategy and set-ups company wide, impacting 100s of teams.
- leading creating a solution for automated mobile app deployment monitoring, working across 3 organizations and multiple teams, creating a solution which prevented bugs from reaching 100s of millions of customers.
- organization changes like team structure, setting new processes, oncall responsibilities, and influencing the collaboration of teams to drive better software quality, higher reliability and faster development speed

Mar 2020 – Aug 2016 — Bloomberg LP, London
Software Engineer

I worked in the Sharing Framework team as a Tech Lead, providing a set of services to enable users to share, access and edit content with users or groups in their network. The system is heavily used and critical to some of the most popular functionalities of the Bloomberg Terminal. I worked in an agile team, owning the full stack of the software.

- I developed the product by working with the business and engineering stakeholders, collaborating cross-team to lead the integration efforts for key features including immutable shares, real time notifications, the permissioning system.
- I architected, designed and implemented systems that handle multi-million requests per day, ensuring their ability to scale and be resilient to failures.
- I was part of the group coordinating *Software Development Life Cycle* improvements for my area (200+ engineers): I especially cared about improving testing, monitoring, and integrating CI&CD.
- I mentored and provided training to team members and interns: I won the *Best mentor of the year* award out of 150+ mentors.
- I was active across the company contributing to Bloomberg-wide internal infrastructure (improvements to build speed, easier testing, fixes and enhancements to the internal C++ standard library).

Publications & Talks

Book **C++ Fundamentals - Francesco Zoffoli, Antonio Mallia**
Packt Publishing, 2019, ISBN: 9781789801491
The book covers the most fundamental aspects of C++ for application development, guiding the reader from beginner topics up to the more advanced ones.
Particular attention is given to modern features introduced in the standards up to C++17.
It sold several thousands copies and had very positive reviews.
Publisher Website

Talk **Coroutine Patterns and How to Use Them**
CppCon 2023
A collections of several recurring patterns when using coroutines in C++, with insightful explanations on when you want to use them and which problems do they solve.
Abstract | Video

Talk **How to Use Dependency Injection to Write Maintainable Software**
CppCon 2022
Show several techniques and tools to use dependency injection in large C++systems. Provides direct, actionable suggestions, with concrete examples in C++20, and dives into less common, yet important, cases.
Abstract | Video

Talk **Automatically Process Your Operations in Bulk With Coroutines**
CppEurope 2022 | CppCon 2021 | C++ London May 2020
Shows how leverage coroutines to perform operations automatically in bulk. It starts explaining how coroutines work in C++20 and builds on top of that. Showing interactive visualizations, it teaches how to use coroutines to perform bulk operations.
Abstract | Video

Education

Apr 2016 – Oct 2013 Master of Science in Computer Engineering, **Politecnico di Milano**, Milan
Score: 110/110 *cum laude*

Feb 2016 – Oct 2015 Internship Researcher at **National University of Singapore**, Singapore
For the Master thesis

Jun 2014 – Jan 2014 Exchange Student at **Case Western Reserve University**, Cleveland, Ohio, USA
GPA: 4.0/4.0

Sept 2013 – Sept 2010 Bachelor of Science in Computer Engineering, **Politecnico di Milano**, Milan
Score: 99/110

Skills

Languages	*Expert*: C++ | *Proficient*: Python, Java, JavaScript
Technologies	RDBMS(MySql, PostgreSQL), Caching (Redis, Memcached), Queuing (RabbitMQ, Kafka), RPC (Thrift), internal solutions similar to public cloud offerings
Tech Skills	software architecture, testing, distributed systems, API design, prompt and context engineering for LLMs
Soft Skills	communication, project management, teamwork, leadership

Tech Interests

Working collaboratively, best practices in software engineering, mentoring and teaching, programming languages, open source, distributed systems, development methodologies

Interests

Skiing, cooking, kite surfing, beach volleyball, opera